

****Modeling Assignment 1: Group Component****

```
In [1]: import numpy as np
import scipy.integrate as spi
import matplotlib.pyplot as plt

print("Packages imported")
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Packages imported

1. Problem Statement

Construct a mathematical model of the concentration of the four greenhouse gases, carbon dioxide CO_2 , methane CH_4 , nitrous oxide N_2O , and water vapour H_2O , as a function of time.

2. Variables and Parameters

Description	Symbol	Dimension	Type
time	t	T	independent variable
concentration of H_2O	C_0	$N\ L^{-3}$	dependent variable
concentration of CO_2	C_1	$N\ L^{-3}$	dependent variable
concentration of CH_4	C_2	$N\ L^{-3}$	dependent variable
concentration of N_2O	C_3	$N\ L^{-3}$	dependent variable
concentration of O_2	C_4	$N\ L^{-3}$	parameter
rate of volcanic eruptions emitting H_2O	R_0	$N\ L^{-3}\ T^{-1}$	parameter
rate of volcanic eruptions emitting CO_2	R_1	$N\ L^{-3}\ T^{-1}$	parameter
rate of wetlands emitting CO_2	R_2	$N\ L^{-3}\ T^{-1}$	parameter
rate of wetlands emitting CH_4	R_3	$N\ L^{-3}\ T^{-1}$	parameter
rate of wetlands emitting N_2O	R_4	$N\ L^{-3}\ T^{-1}$	parameter

rate of water evaporation	R_5	$\text{N L}^{-3} \text{T}^{-1}$	parameter
rate of water precipitation	R_6	$\text{N L}^{-3} \text{T}^{-1}$	parameter
rate of oceans absorbing CO_2	R_7	$\text{N L}^{-3} \text{T}^{-1}$	parameter
rate of oceans absorbing N_2O	R_8	$\text{N L}^{-3} \text{T}^{-1}$	parameter
rate of plant life absorbing CO_2	R_9	$\text{N L}^{-3} \text{T}^{-1}$	parameter
reaction coefficient of CH_4 breaking down	k	$(\text{N L}^{-3})^{-2} \text{T}^{-1}$	parameter
initial concentration of H_2O	$C_{0,0}$	N L^{-3}	parameter
initial concentration of CO_2	$C_{1,0}$	N L^{-3}	parameter
initial concentration of CH_4	$C_{2,0}$	N L^{-3}	parameter
initial concentration of N_2O	$C_{3,0}$	N L^{-3}	parameter

3. Assumptions and Constraints

- There are sufficient O_2 in the atmosphere for the CH_4 to break down.
- The concentration of O_2 in the atmosphere is constant.
- All rates of change in concentration above are constant.
- The four greenhouse gases do not react with each other apart from the CH_4 breakdown.
- The four greenhouse gases are well mixed.
- We know the concentration of O_2 and it is constant.
- We know the initial concentration of the four greenhouse gases.

4. Building Solutions

Apply the mass balance equation to each of the four greenhouse gases:

$$\begin{aligned}
 \frac{dC_0}{dt} &= R_0 + R_5 - R_6 + 2kC_2C_4^2 \\
 \frac{dC_1}{dt} &= R_1 + R_2 - R_7 - R_9 + kC_2C_4^2 \\
 \frac{dC_2}{dt} &= R_3 - kC_2C_4^2 \\
 \frac{dC_3}{dt} &= R_4 - R_8 \\
 C_0(0) &= C_{0,0} \quad C_1(0) = C_{1,0} \quad C_2(0) = C_{2,0} \quad C_3(0) = C_{3,0}
 \end{aligned}$$

Apply the nondimensionalization procedure. Let $C_0 = [C_0]C_0^*$, $C_1 = [C_1]C_1^*$, $C_2 = [C_2]C_2^*$, $C_3 = [C_3]C_3^*$, $C_4 = [C_4]C_4^*$, and $t = [t]t^*$.

Make the substitutions:

$$\frac{[C_0]}{[t]} \frac{dC_0^*}{dt^*} = R_0 + R_5 - R_6 + [C_2][C_4]^2 2kC_2^*C_4^{*2}$$

$$\frac{[C_1]}{[t]} \frac{dC_1^*}{dt^*} = R_1 + R_2 - R_7 - R_9 + [C_2][C_4]^2 kC_2^*C_4^{*2}$$

$$\frac{[C_2]}{[t]} \frac{dC_2^*}{dt^*} = R_3 - [C_2][C_4]^2 kC_2^*C_4^{*2}$$

$$\frac{[C_3]}{[t]} \frac{dC_3^*}{dt^*} = R_4 - R_8$$

$$[C_0]C_0^*(0) = C_{0,0} \quad [C_1]C_1^*(0) = C_{1,0} \quad [C_2]C_2^*(0) = C_{2,0} \quad [C_3]C_3^*(0) = C_{3,0}$$

We choose the same scaling factor for the concentrations

$[C_0] = [C_1] = [C_2] = [C_3] = [C_4] = [C]$ and simplify

$$\frac{dC_0^*}{dt^*} = [t][C]^{-1}(R_0 + R_5 - R_6) + [t][C]^2 2kC_2^*C_4^{*2}$$

$$\frac{dC_1^*}{dt^*} = [t][C]^{-1}(R_1 + R_2 - R_7 - R_9) + [t][C]^2 kC_2^*C_4^{*2}$$

$$\frac{dC_2^*}{dt^*} = [t][C]^{-1}R_3 - [t][C]^2 kC_2^*C_4^{*2}$$

$$\frac{dC_3^*}{dt^*} = [t][C]^{-1}(R_4 - R_8)$$

$$C_0^*(0) = \frac{C_{0,0}}{[C]} \quad C_1^*(0) = \frac{C_{1,0}}{[C]} \quad C_2^*(0) = \frac{C_{2,0}}{[C]} \quad C_3^*(0) = \frac{C_{3,0}}{[C]}$$

From the $\frac{dC_2^*}{dt^*}$ equation, we choose $[C] = (\frac{R_3}{k})^{1/3}$, and then using $[C]$ in the second term the same equation, we obtain

$$[t] = \frac{[C]}{R_3} = (\frac{R_3}{k})^{1/3} R_3^{-1} = k^{-1/3} R_3^{-2/3}.$$

Then we can write the system of equations as

$$\begin{aligned}\frac{dC_0^*}{dt^*} &= \mu_0 + 2C_2^*C_4^{*2} \\ \frac{dC_1^*}{dt^*} &= \mu_1 + C_2^*C_4^{*2} \\ \frac{dC_2^*}{dt^*} &= 1 - C_2^*C_4^{*2} \\ \frac{dC_3^*}{dt^*} &= \mu_3 \\ C_0^*(0) &= \alpha_0 \quad C_1^*(0) = \alpha_1 \quad C_2^*(0) = \alpha_2 \quad C_3^*(0) = \alpha_3\end{aligned}$$

where

$$\begin{aligned}\mu_0 &= \frac{(R_0+R_5-R_6)}{R_3} & \mu_1 &= \frac{(R_1+R_2-R_7-R_9)}{R_3} & \mu_3 &= \frac{(R_4-R_8)}{R_3} & \alpha_0 &= \left(\frac{k}{R_3}\right)^{\frac{1}{3}}C_{0,0} \\ \alpha_1 &= \left(\frac{k}{R_3}\right)^{\frac{1}{3}}C_{1,0} & \alpha_2 &= \left(\frac{k}{R_3}\right)^{\frac{1}{3}}C_{2,0} & \alpha_3 &= \left(\frac{k}{R_3}\right)^{\frac{1}{3}}C_{3,0}.\end{aligned}$$

5. Analyze and Assess

```
In [2]: # constants
mu_0 = 0.001
mu_1 = 0.2
mu_3 = 0.049
c4 = 1.5      # oxygen constant

# initial conditions
c0 = 100      # initial H2O
c1 = 100      # initial CO2
c2 = 100      # initial CH4
c3 = 100      # initial N2O

# DE function
def f(C,t):
    dCdt = np.array([0.,0.,0.,0.,])
    dCdt[0] = mu_0 + 2*C[2]*c4**2
    dCdt[1] = mu_1 + C[2]*c4**2
    dCdt[2] = 1 - C[2]*c4**2
    dCdt[3] = mu_3
    return dCdt

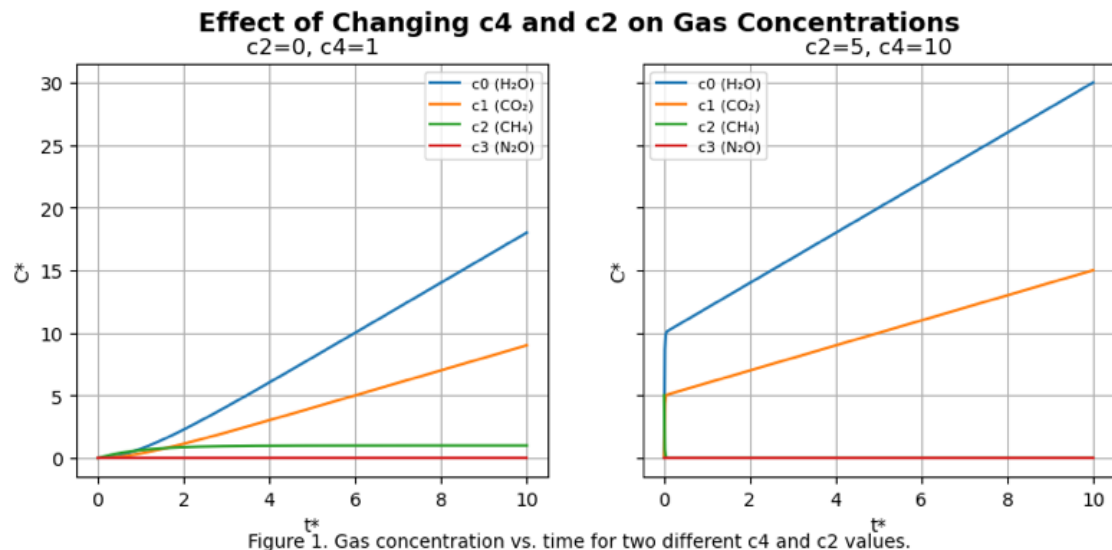
# initial condition
C0 = [c0,c1,c2,c3]
# time vector
t = np.linspace(0,1000,10000) # long term, change values as needed
# DE solution
C = spi.odeint(f,C0,t)

#plot
## uncomment the following lines of code to show graph
```

```
# plt.plot(t,C[:,0],t,C[:,1],t,C[:,2],t,C[:,3])
# plt.grid(True)
# plt.legend(['c0 $H_2O$', 'c1 $CO_2$', 'c2 $CH_4$', 'c3 $N_2O$'])
# plt.xlabel('$t^*$')
# plt.ylabel('$C^*$')
# plt.show()
```

Observations and Analysis

The graph on the left in Figure 1 represents the baseline behaviour of each gas concentration. We set the net concentration for all four greenhouse gases equal to 0 while keeping a constant nonzero oxygen (O_2 ; c_4) concentration such that the breakdown of methane could still happen. We see that methane (CH_4) and nitrous oxide (N_2O) concentration have a steady state close to 0, while water vapour (H_2O) and carbon dioxide (CO_2) show a linearly increasing trend. When we increased the initial condition value of O_2 and CH_4 concentration (c_2 ; green line), the concentration for both H_2O and CO_2 increased more rapidly in the beginning. We observed that the concentration of CH_4 reached a lower steady state when the initial values of concentration of CH_4 and O_2 were higher. These findings indicate that CH_4 (combined with O_2) breaks down into CO_2 and H_2O . As a result, when more CH_4 and O_2 are present in the atmosphere, the breakdown happens more rapidly leading to a higher concentration of CO_2 and H_2O .



In the model, each μ represents the net balance between emission and absorption of the gases. Changing these μ values (μ_0, μ_1, μ_3) corresponding to H_2O , CO_2 , and N_2O , respectively, seemed to affect the slope of their concentration curves. For instance, larger μ values produced a steeper linear

graph, suggesting a stronger net source over time. When plotting many different graphs with alternated initial conditions, we observed that different initial conditions changed the starting values of the lines, while the slope remained the same. This aligns with our deterministic model, because $\frac{dC_3}{dt} = \mu_3$ is constant, allowing its slope to solely depend on μ_3 .

The concentration rates for CH_4 and N_2O appeared to be less susceptible to change compared to those for H_2O and CO_2 . Again, this observation aligns with our model, as the rates of CH_4 and N_2O depend only on constant parameters (c_4 and μ_3) so they are less sensitive to other sources (i.e., evaporation and precipitation) compared to H_2O and CO_2 .

The graph on the left of Figure 2 shows a short term behaviour ($t \in [0, 10]$), while that on the right shows a long term behaviour ($t \in [0, 1000]$). When we looked at these curves in the short term, they all seemed to come to certain steady states after an initial curve (either a decay or growth). However, when looked at in the long run, we observed that the three gases (H_2O , CO_2 , and N_2O) still increased linearly, with CH_4 still coming to a steady state of 0.

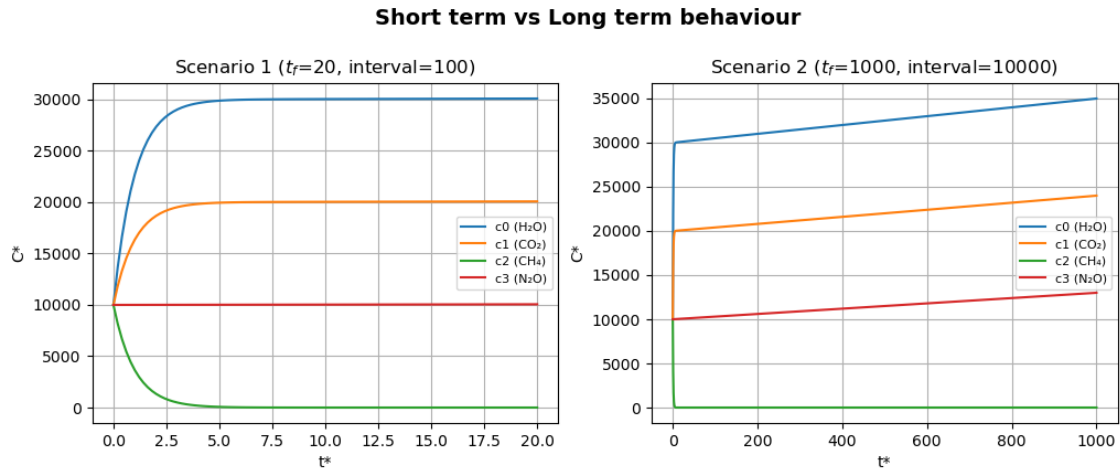


Figure 2. Short term figure on the left; long term figure on the right

Model Limitations

Since we assumed the volcanic eruption released the greenhouse gases at a constant rate, the lines of H_2O and CO_2 in the plots looked smoother than the real case. We also assumed that all rates of emission and absorption of greenhouse gases were at constant rates, which erased the fluctuation of the concentration lines. Furthermore, the sources that emit greenhouse gases might not keep emitting gases forever, so our model only gives information for the concentrations in the foreseeable future.